/ 1.75

(1) LaTeX-Bausteine im Überblick

Lösungen

Lösungstext

```
\Loesung{text}
```

text

Im Erwartungshorizont ersetzen

```
\LoesungReplace{solution}{blankContent}
```

solution

Leerraum

```
\LoesungLeer{solution}{2cm}
```

solution

Lösungsbild

```
\LoesungImg{444.jpg}{100.jpg}{5cm}
```



Lücke zum Ausfüllen

```
\LoesungLuecke{solution}{3cm}
```

solution

Karopapier

```
\LoesungKaro{solution}{3}
```

solution

Linierte Zeilen

```
\LoesungLine{solution}{3}
```

solution

Karopapier (ohne Lösung)

```
\kariert[0.5cm]{3}
```

Linierte Zeilen (ohne Lösung)

```
\liniert[0.8cm]{3}
```

Punkte

Ganzer Punkt

```
\BE
```

Halber Punkt

```
\hBE
```

Viertelpunkt

```
\qBE
```

Teilaufgabe

```
\begin{Teilaufgabe}{Teilaufgabe}
\end{Teilaufgabe}
```

a) Teilaufgabe

Formatierung

Fetter Text

```
\textbf{text}
```

text

Kursiver Text

```
\textit{text}
```

text

Farbiger Text

```
\textcolor{red}{text}
```

text

Kleiner Text

```
{\footnotesize text}
```

text

Etwas größerer Text

```
{\large text}
```

text

Größerer Text

```
{\Large text}
```

text

Große Überschrift

```
{\LARGE text}
```

text

Sehr großer Text

```
{\huge text}
```

text

Allgemein

Bild

```
\includegraphics[width=8cm]{200.jpg}
```



Tabelle

```
\begin{tabular}{ll}
  a & b \\
  c & d \\
\end{tabular}
```

a	b
c	d

Aufzählung

```
\begin{itemize}
  \item item
  \item item
\end{itemize}
```

- item
- item

Nummerierte Liste

```
\begin{enumerate}
  \item item
  \item item
\end{enumerate}
```

1. item
2. item

Formel im Text

```
$x$
```

x

Zentrierter Inhalt

```
\begin{center}
  content
\end{center}
```

content

Seitenumbruch

```
\Seitenumbruch
```

(erzwingt hier einen Seitenumbruch; im Überblick nicht ausgeführt)

Mathematik

Formel

```
$a^2 + b^2 = c^2$
```

$a^2 + b^2 = c^2$

$\mathbb{N}$ natürliche Zahlen

```
$\mathbb{N}$
```

$\mathbb{N}$

$\mathbb{Z}$ ganze Zahlen

```
$\mathbb{Z}$
```

$\mathbb{Z}$

$\mathbb{Q}$ rationale Zahlen

```
$\mathbb{Q}$
```

$\mathbb{Q}$

$\mathbb{R}$ reelle Zahlen

```
$\mathbb{R}$
```

$\mathbb{R}$

$\mathbb{C}$ komplexe Zahlen

```
$\mathbb{C}$
```

$\mathbb{C}$

Bruch

```
$\frac{a}{b}$
```

$\frac{a}{b}$

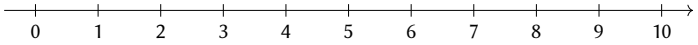
Wurzel

```
$\sqrt{x}$
```

$\sqrt{x}$

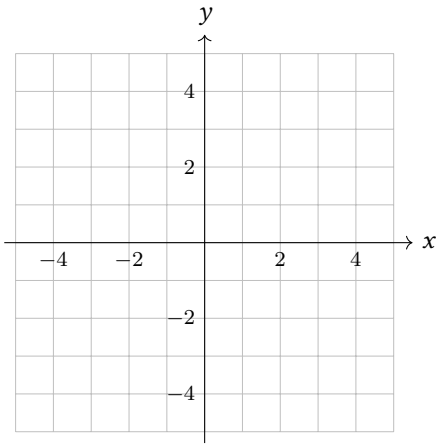
Zahlenstrahl

```
\begin{tikzpicture}
  % Zahlenstrahl mit Pfeilspitze
  \draw[->] (-0.5,0) -- (10.5,0);
  % Teilstriche und Zahlen
  \foreach \x in {0,...,10} \draw (\x,0.1) -- (\x,-0.1)
    node[below] {\x};
\end{tikzpicture}
```



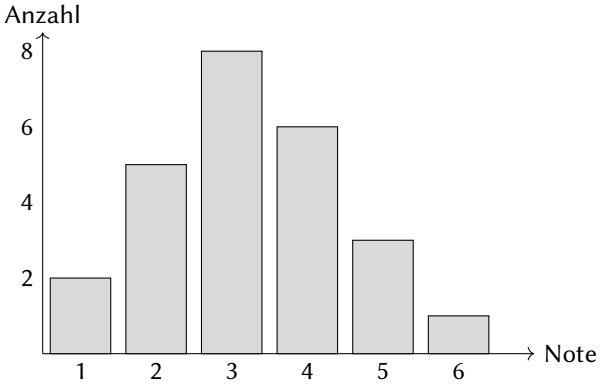
Koordinatengitter

```
\begin{tikzpicture}[scale=0.5]
  % Koordinatengitter
  \draw[very thin, gray!50] (-5,-5) grid (5,5);
  % Achsen
  \draw[->] (-5.3,0) -- (5.5,0) node[right] {$x$};
  \draw[->] (0,-5.3) -- (0,5.5) node[above] {$y$};
  % Achsenbeschriftung
  \foreach \x in {-4,-2,2,4} \node[below] at (\x,0) {
    \footnotesize $\x$};
  \foreach \y in {-4,-2,2,4} \node[left] at (0,\y) {
    \footnotesize $\y$};
\end{tikzpicture}
```



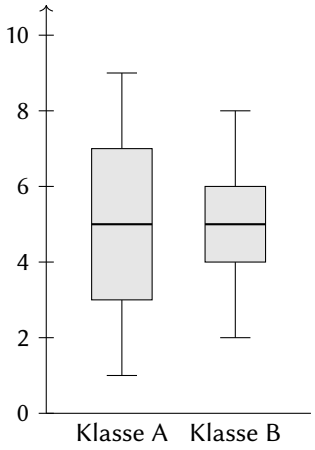
Säulendiagramm

```
\begin{tikzpicture}[yscale=0.5]
  % Achsen
  \draw[->] (0,0) -- (6.5,0) node[right] {Note};
  \draw[->] (0,0) -- (0,8.5) node[above] {Anzahl};
  % Datenbalken (Note / Haeufigkeit)
  \foreach \n/\h in {1/2, 2/5, 3/8, 4/6, 5/3, 6/1}
    \draw[fill=gray!30] ({\n-0.9},0) rectangle ({\n-0.1},
    \h);
  % Beschriftung
  \foreach \n in {1,...,6} \node[below] at ({\n-0.5},0) {
    \n};
  \foreach \y in {2,4,6,8} \node[left] at (0,\y) {\y};
\end{tikzpicture}
```



Boxplot

```
\begin{tikzpicture}[yscale=0.5]
  % Skala (y-Achse)
  \draw[->] (0,0) -- (0,10.8);
  \foreach \y in {0,2,...,10} \draw (0.1,\y) -- (-0.1,\y)
     node[left] {\y};
  \draw (0,0) -- (3.5,0);
  % Boxplots: x / Beschriftung / Min / Q1 / Median / Q3 /
     Max
  \foreach \x/\l/\mn/\qa/\me/\qb/\mx in {1/Klasse A
    /1/3/5/7/9, 2.5/Klasse B/2/4/5/6/8} {
    % Antennen (Whiskers)
    \draw (\x,\mn) -- (\x,\qa) (\x,\qb) -- (\x,\mx);
    \draw (\x-0.2,\mn) -- (\x+0.2,\mn) (\x-0.2,\mx) -- (
     \x+0.2,\mx);
    % Box (Interquartilsabstand)
    \draw[fill=gray!20] (\x-0.4,\qa) rectangle (\x+0.4,
     \qb);
    % Medianlinie
    \draw[thick] (\x-0.4,\me) -- (\x+0.4,\me);
    \node[below] at (\x,0) {\l};
  }
\end{tikzpicture}
```



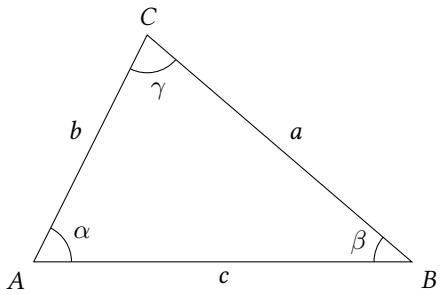
Gleichungssystem

```
\begin{align*}
  2x + 3y &= 7 \\
  x - y &= 1
\end{align*}
```

$$2x + 3y = 7$$
$$x - y = 1$$

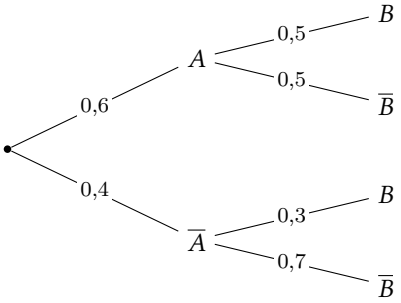
Dreieck

```
\begin{tikzpicture}
  % Eckpunkte
  \coordinate[label=below left:$A$] (A) at (0,0);
  \coordinate[label=below right:$B$] (B) at (5,0);
  \coordinate[label=above:$C$] (C) at (1.5,3);
  % Dreiecksseiten
  \draw (A) -- node[below] {$c$} (B) -- node[above right]
     {$a$} (C) -- node[above left] {$b$} (A);
  % Innenwinkel
  \pic[draw, "$\alpha$", angle eccentricity=1.5] {angle =
     B--A--C};
  \pic[draw, "$\beta$", angle eccentricity=1.5] {angle =
     C--B--A};
  \pic[draw, "$\gamma$", angle eccentricity=1.5] {angle =
     A--C--B};
\end{tikzpicture}
```



Baumdiagramm

```
\begin{tikzpicture}[grow'=right, level distance=2.5cm,
    level 1/.style={sibling distance=2.4cm},
    level 2/.style={sibling distance=1.2cm}, prob/.style
    ={fill=white, inner sep=1pt, font=\footnotesize}]
  % Wurzel und 1. Stufe
  \node[circle, fill, inner sep=1pt] {}
    child {node {$A$}
      % 2. Stufe fuer Pfad A
      child {node {$B$} edge from parent node[prob]
     {$0{,}5$}}
      child {node {$\overline{B}$} edge from parent node[
     prob] {$0{,}5$}}
      edge from parent node[prob] {$0{,}6$}}
    child {node {$\overline{A}$}
      % 2. Stufe fuer Pfad nicht A
      child {node {$B$} edge from parent node[prob]
     {$0{,}3$}}
      child {node {$\overline{B}$} edge from parent node[
     prob] {$0{,}7$}}
      edge from parent node[prob] {$0{,}4$}};
\end{tikzpicture}
```



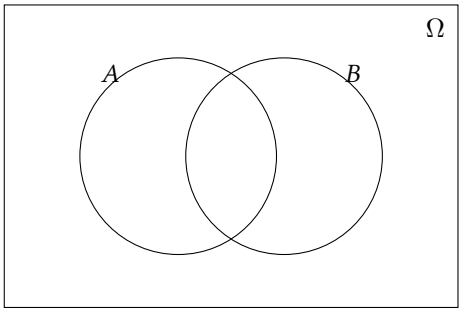
Vierfeldertafel

```
\begin{tabular}{c|c|c|c}
   & $B$ & $\overline{B}$ & \\ \hline
  $A$ & $0{,}2$ & & \\ \hline
  $\overline{A}$ & & & \\ \hline
   & & & $1$
\end{tabular}
```

	B	$\overline{B}$	
A	$0{,}2$		
$\overline{A}$			
			1

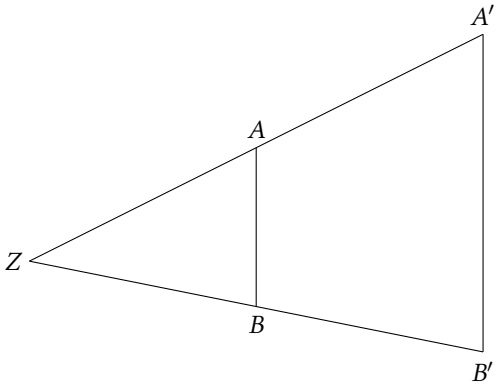
Mengendiagramm

```
\begin{tikzpicture}
  % Grundmenge Omega
  \draw (0,0) rectangle (6,4);
  \node at (5.7,3.7) {$\Omega$};
  % Mengen A und B
  \draw (2.3,2) circle (1.3);
  \draw (3.7,2) circle (1.3);
  \node at (1.4,3.1) {$A$};
  \node at (4.6,3.1) {$B$};
\end{tikzpicture}
```



Strahlensatzfigur

```
\begin{tikzpicture}
  % Scheitelpunkt und Punkte
  \coordinate[label=left:$Z$] (Z) at (0,0);
  \coordinate[label=above:$A$] (A) at (3,1.5);
  \coordinate[label=below:$B$] (B) at (3,-0.6);
  \coordinate[label=above:$A'$] (A2) at (6,3);
  \coordinate[label=below:$B'$] (B2) at (6,-1.2);
  % Strahlen
  \draw (Z) -- (A2);
  \draw (Z) -- (B2);
  % Parallelen
  \draw (A) -- (B);
  \draw (A2) -- (B2);
\end{tikzpicture}
```



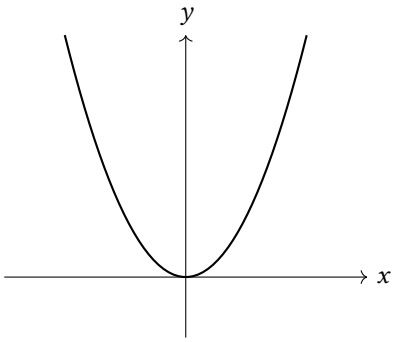
Wertetabelle

```
\begin{tabular}{|c|c|c|c|c|}
  \hline
  $x$ & 0 & 1 & 2 & 3 \\ \hline
  $f(x)$ & & & & & \\ \hline
\end{tabular}
```

x	0	1	2	3
$f(x)$				

Funktionsgraph

```
\begin{tikzpicture}[scale=0.8]
  % Koordinatenachsen
  \draw[->] (-3,0) -- (3,0) node[right] {$x$};
  \draw[->] (0,-1) -- (0,4) node[above] {$y$};
  % Funktionsgraph
  \draw[thick, domain=-2:2, samples=50] plot (\x, {(\x)
    ^2});
\end{tikzpicture}
```



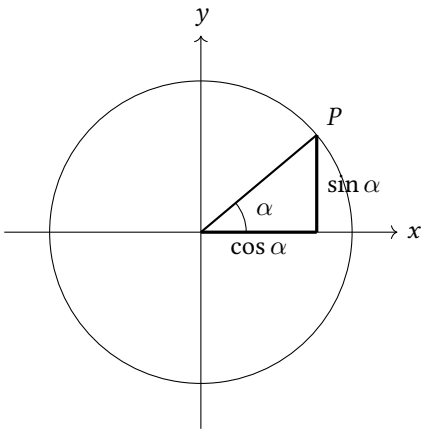
Fallunterscheidung

```
$f(x) = \begin{cases} x^2 & x \geq 0 \\ -x & x < 0 \end{
    cases}$
```

$$f(x) = \begin{cases} x^2 & x \geq 0 \\ -x & x < 0 \end{cases}$$

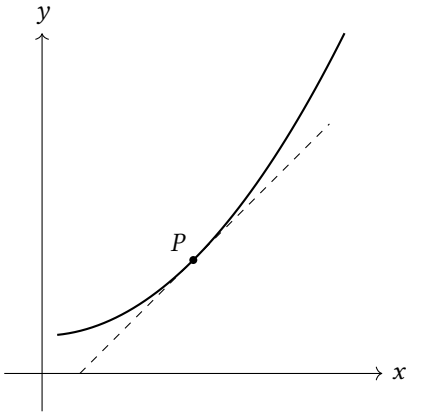
Einheitskreis

```
\begin{tikzpicture}[scale=2]
  % Achsen und Einheitskreis
  \draw[->] (-1.3,0) -- (1.3,0) node[right] {$x$};
  \draw[->] (0,-1.3) -- (0,1.3) node[above] {$y$};
  \draw (0,0) circle (1);
  % Zeiger und Punkt P
  \draw[thick] (0,0) -- (40:1) node[above right] {$P$};
  % Kosinus- und Sinus-Strecke
  \draw[very thick] (0,0) -- ({cos(40)},0) node[midway,
    below] {$\cos\alpha$};
  \draw[very thick] ({cos(40)},0) -- (40:1) node[midway,
    right] {$\sin\alpha$};
  % Winkelbogen
  \draw (0.3,0) arc (0:40:0.3);
  \node at (20:0.45) {$\alpha$};
\end{tikzpicture}
```



Tangente

```
\begin{tikzpicture}
  % Koordinatenachsen
  \draw[->] (-0.5,0) -- (4.5,0) node[right] {$x$};
  \draw[->] (0,-0.5) -- (0,4.5) node[above] {$y$};
  % Funktionsgraph
  \draw[thick, domain=0.2:4, samples=50] plot (\x,
    {0.25*(\x)^2+0.5});
  % Tangente und Beruehrpunkt P
  \draw[dashed, domain=0.5:3.8] plot (\x, {\x-0.5});
  \fill (2,1.5) circle (1.5pt) node[above left] {$P$};
\end{tikzpicture}
```



Matrix

```
$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$
```

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

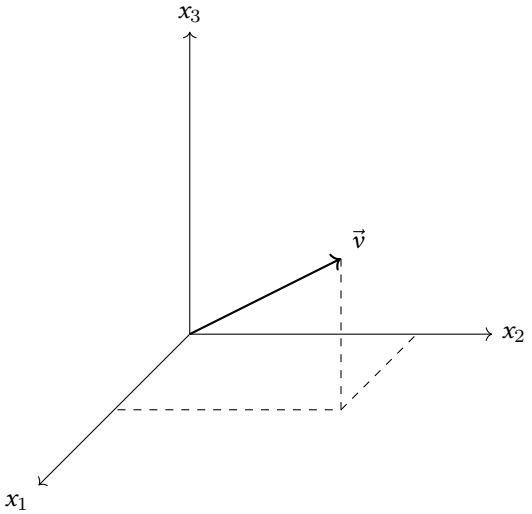
Spaltenvektor

```
$\vec{v} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$
```

$$\vec{v} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$

3D-Koordinatensystem

```
\begin{tikzpicture}[x={(-0.5cm,-0.5cm)}, y={(1cm,0cm)}, z
    ={(0cm,1cm)}]
  % 3D-Koordinatenachsen
  \draw[->] (0,0,0) -- (4,0,0) node[below left] {$x_1$};
  \draw[->] (0,0,0) -- (0,4,0) node[right] {$x_2$};
  \draw[->] (0,0,0) -- (0,0,4) node[above] {$x_3$};
  % Vektorpfeil und Projektionslinien
  \draw[thick, ->] (0,0,0) -- (2,3,2) node[above right]
    {$\vec{v}$};
  \draw[dashed] (2,3,2) -- (2,3,0) -- (2,0,0);
  \draw[dashed] (2,3,0) -- (0,3,0);
\end{tikzpicture}
```



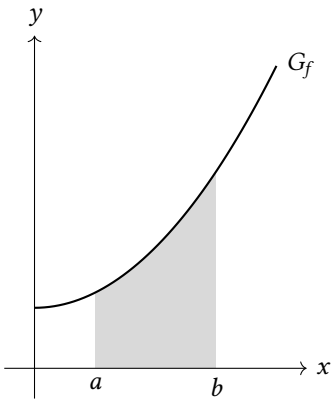
Integral

```
$\int_{a}^{b} f(x)\,\mathrm{d}x$
```

$$\int_a^b f(x)\,\mathrm{d}x$$

Fläche unter dem Graphen

```
\begin{tikzpicture}[scale=0.8]
  % Flaeche unter dem Graphen
  \fill[gray!30, domain=1:3, samples=40] (1,0) -- plot (
    \x, {0.25*(\x)^2+1}) -- (3,0) -- cycle;
  % Achsen
  \draw[->] (-0.5,0) -- (4.5,0) node[right] {$x$};
  \draw[->] (0,-0.5) -- (0,5.5) node[above] {$y$};
  % Funktionsgraph
  \draw[thick, domain=0:4, samples=50] plot (\x, {0.25*(
    \x)^2+1}) node[right] {$G_f$};
  % Integrationsgrenzen
  \node[below] at (1,0) {$a$};
  \node[below] at (3,0) {$b$};
\end{tikzpicture}
```



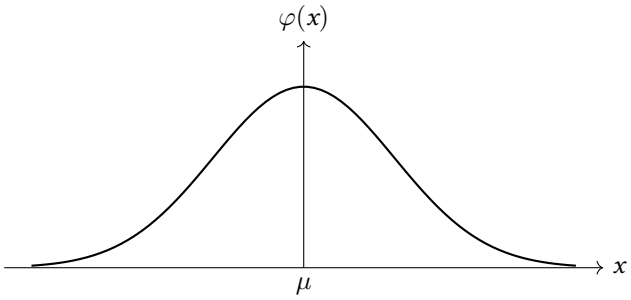
Summe

```
$\sum_{i=1}^{n} i$
```

$\sum_{i=1}^{n} i$

Normalverteilung

```
\begin{tikzpicture}[xscale=1.2, yscale=6]
  % Koordinatenachsen
  \draw[->] (-3.3,0) -- (3.3,0) node[right] {$x$};
  \draw[->] (0,0) -- (0,0.5) node[above] {$\varphi(x)$};
  % Gauss-Glockenkurve
  \draw[thick, domain=-3:3, samples=80] plot (\x, {exp(-(
    \x)^2/2)/sqrt(2*pi)});
  % Erwartungswert mu
  \node[below] at (0,0) {$\mu$};
\end{tikzpicture}
```



Informatik

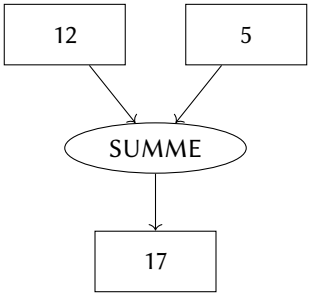
Tabellenblatt

```
\begin{tabular}{|c|c|c|c|}
  \hline
   & A & B & C \\ \hline
  1 & Artikel & Preis & Menge \\ \hline
  2 & Cola & 1,50 & 3 \\ \hline
  3 & & & \texttt{=B2*C2} \\ \hline
\end{tabular}
```

	A	B	C
1	Artikel	Preis	Menge
2	Cola	1,50	3
3			=B2*C2

Datenflussdiagramm

```
\begin{tikzpicture}[data/.style={draw, minimum width=1.6
    cm, minimum height=0.8cm},
  func/.style={draw, ellipse, minimum width=2.4cm}]
  % Datenknoten
  \node[data] (a) at (-1.2,3) {12};
  \node[data] (b) at (1.2,3) {5};
  % Funktionsknoten
  \node[func] (f) at (0,1.5) {SUMME};
  % Ergebnis
  \node[data] (r) at (0,0) {17};
  % Datenfluesse
  \draw[->] (a) -- (f);
  \draw[->] (b) -- (f);
  \draw[->] (f) -- (r);
\end{tikzpicture}
```



Relationenschema

```
\texttt{Schueler}(\underline{id : int}, name : String,
    \dotuline{klasse\_id : int})
```

Schueler(id : int, name : String, klasse_id : int)

Datenbanktabelle

```
\begin{tabular}{c | c | c}
id & name & alter \\ \hline\hline
1 & "Lia" & 15 \\ \hline
2 & "Ben" & 16
\end{tabular}
```

id	name	alter
1	”Lia”	15
2	”Ben”	16

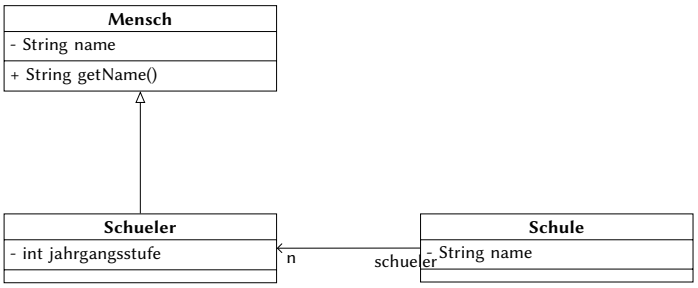
SQL-Lösung

```
\LoesungLine{SELECT name FROM Schueler WHERE alter >
    15}{2}
```

SELECT name FROM Schueler WHERE alter > 15

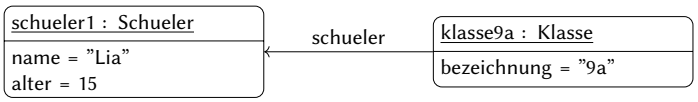
Klassendiagramm

```
\begin{tikzpicture}
  % Oberklasse
  \begin{class}[text width=5cm]{Mensch}{0,0}
    \attribute{- String name}
    \operation{+ String getName()}
  \end{class}
  % Unterklasse mit Vererbung
  \begin{class}[text width=5cm]{Schueler}{0,-4}
    \inherit{Mensch}
    \attribute{- int jahrgangsstufe}
  \end{class}
  % Assoziierte Klasse
  \begin{class}[text width=5cm]{Schule}{8,-4}
    \attribute{- String name}
  \end{class}
  % Assoziation
  \unidirectionalAssociation{Schule}{schueler}{n}{
    Schueler}{}{}
\end{tikzpicture}
```



Objektdiagramm

```
\begin{tikzpicture}[object/.style={draw, rounded corners,
     rectangle split, rectangle split parts=2, text width
    =4cm}]
  % Objekte (Name : Klasse und Attribute)
  \node[object] (s) at (0,0) {\underline{schueler1 :
    Schueler} \nodepart{two} name = "Lia" \\ alter = 15};
  \node[object] (k) at (7,0) {\underline{klasse9a :
    Klasse} \nodepart{two} bezeichnung = "9a"};
  % Objektbeziehung
  \draw[->] (k) -- node[above] {schueler} (s);
\end{tikzpicture}
```



Quellcode

```
\begin{framed}
\begin{lstlisting}[language=Java]
public class Schueler {
    private String name;

    public String getName() {
        return name;
    }
}
\end{lstlisting}
\end{framed}
```

```java
public class Schueler {
    private String name;

    public String getName() {
        return name;
    }
}
```

Lückencode

```
\begin{framed}
\begin{lstlisting}
public class Schueler {

    private (*@\LoesungLuecke{int}{3cm}@*) alter;

    private (*@\LoesungLuecke{String}{3cm}@*) name;
}
\end{lstlisting}
\end{framed}
```

```java
public class Schueler {

    private int alter;

    private String name;
}
```

Code nebeneinander

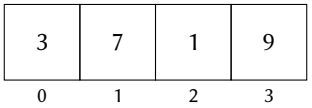
```
\noindent
\begin{minipage}[t]{0.48\textwidth}
\begin{framed}
\begin{lstlisting}
public class Mensch {
    private String name;
}
\end{lstlisting}
\end{framed}
\end{minipage}
\hfill
\begin{minipage}[t]{0.48\textwidth}
\begin{framed}
\begin{lstlisting}
public class Schueler extends Mensch {
    private int jahrgangsstufe;
}
\end{lstlisting}
\end{framed}
\end{minipage}
```

```
public class
    Mensch {
     private
         String
         name;
}
```

```
public class
    Schueler
    extends Mensch
     {
     private int
         jahrgangsstufe
         ;
}
```

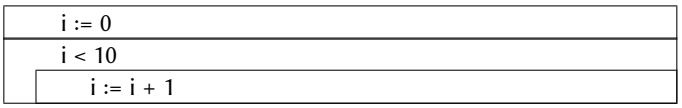
Array

```
\begin{tikzpicture}
  % Array-Zellen: Wert oben, Index unten
  \foreach \v [count=\i from 0] in {3, 7, 1, 9} {
    \draw (\i,0) rectangle ++(1,1);
    \node at (\i+0.5,0.5) {\v};
    \node[below] at (\i+0.5,0) {\footnotesize \i};
  }
\end{tikzpicture}
```



Struktogramm

```
\begin{struktogramm}(90,40)
  % Anweisung
  \assign{i := 0}
  % Wiederholungsstruktur
  \while{i < 10}
    \assign{i := i + 1}
  \whileend
\end{struktogramm}
```



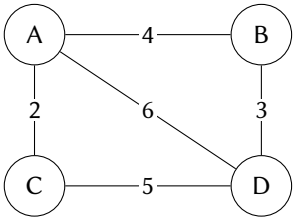
ER-Diagramm

```
\begin{tikzpicture}[entity/.style={draw, minimum width
    =2.5cm, minimum height=1cm},
  relation/.style={draw, diamond, aspect=2}]
  % Entitaeten
  \node[entity] (a) {Lehrer};
  \node[relation, right=1.5cm of a] (r) {unterrichtet};
  \node[entity, right=1.5cm of r] (b) {Klasse};
  % Beziehung mit Kardinalitaeten
  \draw (a) -- node[above] {n} (r) -- node[above] {m} (b)
    ;
\end{tikzpicture}
```



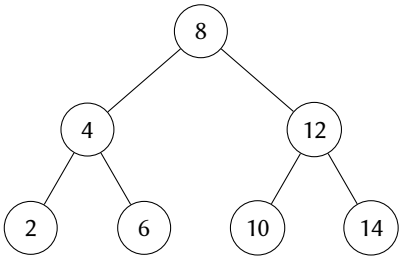
Graph

```
\begin{tikzpicture}[vertex/.style={circle, draw, minimum
    size=8mm}, w/.style={fill=white, inner sep=1pt}]
  % Knoten
  \node[vertex] (A) at (0,2) {A};
  \node[vertex] (B) at (3,2) {B};
  \node[vertex] (C) at (0,0) {C};
  \node[vertex] (D) at (3,0) {D};
  % Gewichtete Kanten
  \draw (A) -- node[w] {4} (B);
  \draw (A) -- node[w] {2} (C);
  \draw (B) -- node[w] {3} (D);
  \draw (C) -- node[w] {5} (D);
  \draw (A) -- node[w] {6} (D);
\end{tikzpicture}
```



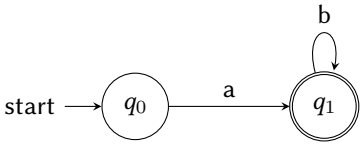
Binärbaum

```
\begin{tikzpicture}[level distance=1.3cm, level 1/.style
    ={sibling distance=3cm},
    level 2/.style={sibling distance=1.5cm}, every node/.
    style={circle, draw, minimum size=7mm}]
  % Wurzel und Teilbaeume
  \node {8}
    child {node {4} child {node {2}} child {node {6}}}
    child {node {12} child {node {10}} child {node
    {14}}};
\end{tikzpicture}
```



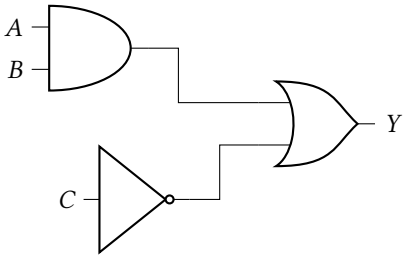
Automat

```
\begin{tikzpicture}[->, >=stealth, node distance=2.5cm,
    auto]
  % Zustaende (Start- und Endzustand)
  \node[state, initial] (q0) {$q_0$};
  \node[state, accepting, right of=q0] (q1) {$q_1$};
  % Uebergaenge
  \path (q0) edge node {a} (q1)
        (q1) edge[loop above] node {b} (q1);
\end{tikzpicture}
```



Logikgatter

```
\begin{circuitikz}
  % Logikgatter
  \draw (0,0) node[and port] (g1) {} (0,-2) node[not port
    ] (g2) {} (3,-1) node[or port] (g3) {};
  % Verbindungsleitungen
  \draw (g1.out) -- ++(0.4,0) |- (g3.in 1) (g2.out) --
    ++(0.4,0) |- (g3.in 2);
  % Eingaenge und Ausgang
  \draw (g1.in 1) node[left] {$A$} (g1.in 2) node[left]
    {$B$} (g2.in) node[left] {$C$};
  \draw (g3.out) node[right] {$Y$};
\end{circuitikz}
```



Physik

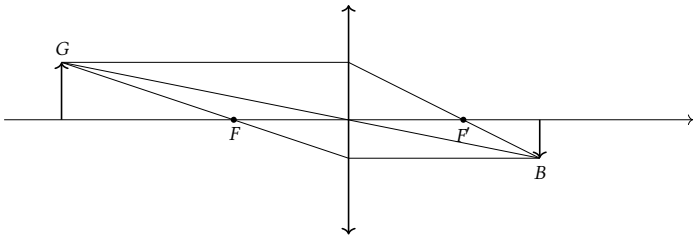
Größe mit Einheit

```
$9{,}81\,\mathrm{m/s^2}$
```

$9{,}81\,\mathrm{m/s^2}$

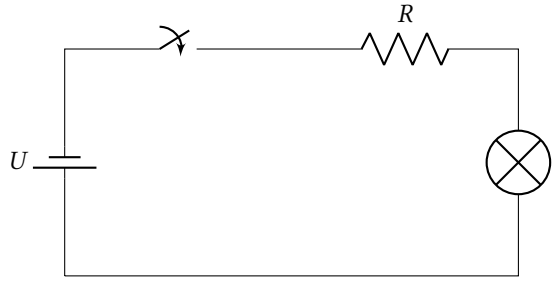
Sammellinse

```
\begin{tikzpicture}
  % Optische Achse und Sammellinse
  \draw[->] (-6,0) -- (6,0);
  \draw[thick, <->] (0,-2) -- (0,2);
  % Brennpunkte
  \fill (-2,0) circle (1.5pt) node[below] {$F$};
  \fill (2,0) circle (1.5pt) node[below] {$F'$};
  % Gegenstandspfeil
  \draw[->, thick] (-5,0) -- (-5,1) node[above] {$G$};
  % Strahlenkonstruktion
  \draw (-5,1) -- (0,1) -- (3.33,-0.67);
  \draw (-5,1) -- (3.33,-0.67);
  \draw (-5,1) -- (0,-0.67) -- (3.33,-0.67);
  % Bildpfeil
  \draw[->, thick] (3.33,0) -- (3.33,-0.67) node[below]
    {$B$};
\end{tikzpicture}
```



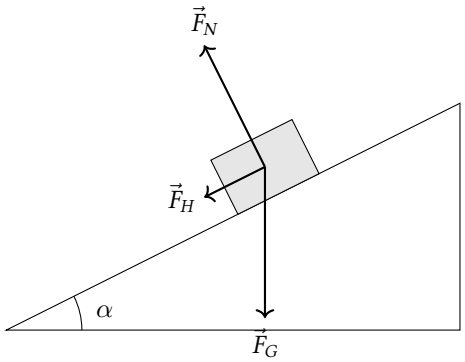
Schaltplan

```
\begin{circuitikz}
  % Stromkreis mit Spannungsquelle, Schalter, Widerstand
    und Lampe
  \draw (0,0) to[battery1, l=$U$] (0,3)
        to[switch] (3,3)
        to[R, l=$R$] (6,3)
        to[lamp] (6,0) -- (0,0);
\end{circuitikz}
```



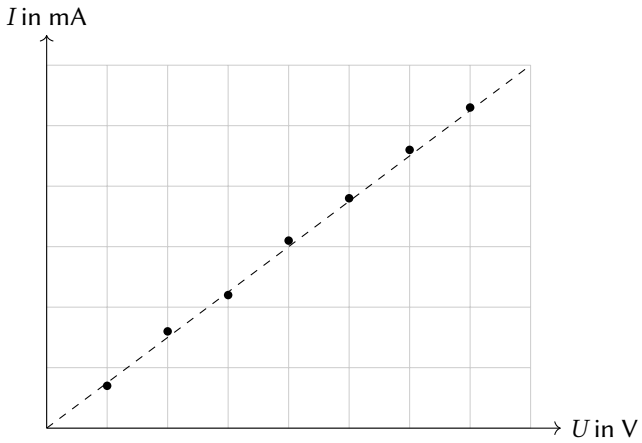
Schiefe Ebene

```
\begin{tikzpicture}
  % Schiefe Ebene
  \draw (0,0) -- (6,0) -- (6,3) -- cycle;
  % Koerper auf der Ebene
  \begin{scope}[shift={(3.6,1.8)}, rotate=26.57]
    \draw[fill=gray!20] (-0.6,0) rectangle (0.6,0.8);
    \coordinate (S) at (0,0.4);
  \end{scope}
  % Kraefte (Gewichtskraft, Normalkraft,
    Hangabtriebskraft)
  \draw[->, thick] (S) -- ++(0,-2) node[below] {$\vec F_G
    $};
  \draw[->, thick] (S) -- ++(116.57:1.8) node[above] {$
    \vec F_N$};
  \draw[->, thick] (S) -- ++(206.57:0.9) node[left] {$
    \vec F_H$};
  % Neigungswinkel
  \draw (1,0) arc (0:26.57:1);
  \node at (1.3,0.25) {$\alpha$};
\end{tikzpicture}
```



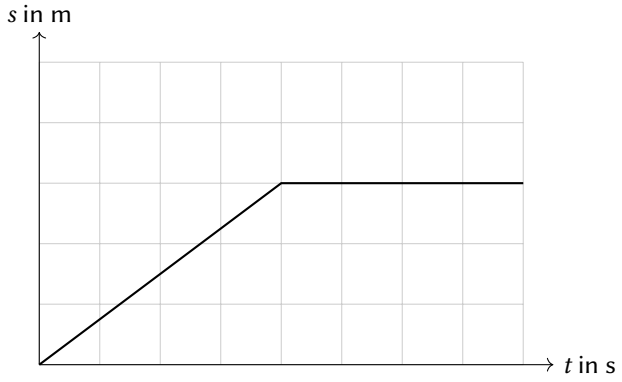
Messwertdiagramm

```
\begin{tikzpicture}[scale=0.8]
  % Achsen und Gitter
  \draw[very thin, gray!40] (0,0) grid (8,6);
  \draw[->] (0,0) -- (8.5,0) node[right] {$U$ in V};
  \draw[->] (0,0) -- (0,6.5) node[above] {$I$ in mA};
  % Messpunkte
  \foreach \x/\y in {1/0.7, 2/1.6, 3/2.2, 4/3.1, 5/3.8,
    6/4.6, 7/5.3}
    \fill (\x,\y) circle (2pt);
  % Ausgleichsgerade
  \draw[dashed] (0,0) -- (8,6);
\end{tikzpicture}
```



t-s-Diagramm

```
\begin{tikzpicture}[scale=0.8]
  % Achsen und Gitter
  \draw[very thin, gray!40] (0,0) grid (8,5);
  \draw[->] (0,0) -- (8.5,0) node[right] {$t$ in s};
  \draw[->] (0,0) -- (0,5.5) node[above] {$s$ in m};
  % Bewegungsverlauf
  \draw[thick] (0,0) -- (4,3) -- (8,3);
\end{tikzpicture}
```



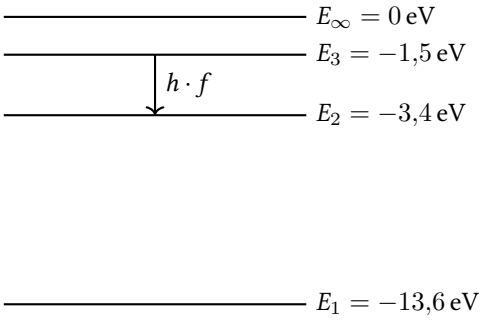
Energieflussdiagramm

```
\begin{tikzpicture}[box/.style={draw, minimum width=2.8cm
    , minimum height=1cm, align=center}]
  % Energieformen und Wandler
  \node[box] (in) at (0,0) {elektrische\\Energie};
  \node[box] (dev) at (4,0) {Glühlampe};
  \node[box] (use) at (8,0.8) {Strahlungs-\\energie};
  \node[box] (loss) at (8,-0.8) {innere\\Energie};
  % Energiefluesse
  \draw[->, very thick] (in) -- (dev);
  \draw[->, very thick] (dev) -- (use);
  \draw[->, thick] (dev) -- (loss);
\end{tikzpicture}
```



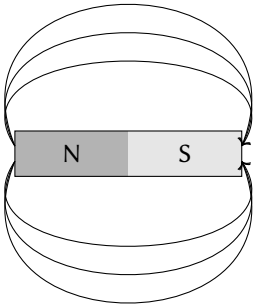
Energieniveauschema

```
\begin{tikzpicture}
  % Energieniveaus
  \draw[thick] (0,0) -- (4,0) node[right] {$E_1 = -13{,}6
    \,\mathrm{eV}$};
  \draw[thick] (0,2.5) -- (4,2.5) node[right] {$E_2 =
    -3{,}4\,\mathrm{eV}$};
  \draw[thick] (0,3.3) -- (4,3.3) node[right] {$E_3 =
    -1{,}5\,\mathrm{eV}$};
  \draw[thick] (0,3.8) -- (4,3.8) node[right] {$E_\infty
    = 0\,\mathrm{eV}$};
  % Uebergang / Photon-Emission
  \draw[->, thick] (2,3.3) -- (2,2.5) node[midway, right]
     {$h \cdot f$};
\end{tikzpicture}
```



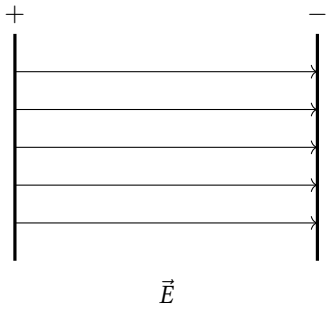
Feldlinien Stabmagnet

```
\begin{tikzpicture}
  % Stabmagnet
  \fill[gray!60] (-1.5,-0.3) rectangle (0,0.3);
  \fill[gray!20] (0,-0.3) rectangle (1.5,0.3);
  \draw (-1.5,-0.3) rectangle (1.5,0.3);
  \node at (-0.75,0) {N};
  \node at (0.75,0) {S};
  % Magnetische Feldlinien
  \foreach \h in {0.6,1.1,1.6} {
    \draw[->] (-1.5,0.1) .. controls (-2.5,\h+1) and
     (2.5,\h+1) .. (1.5,0.1);
    \draw[->] (-1.5,-0.1) .. controls (-2.5,-\h-1) and
     (2.5,-\h-1) .. (1.5,-0.1);
  }
\end{tikzpicture}
```



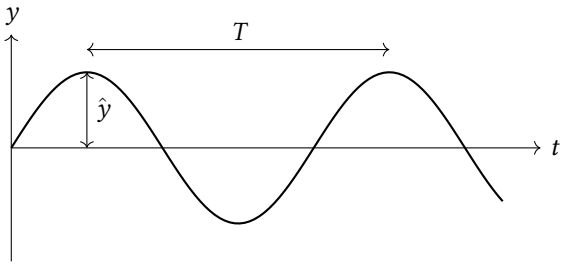
Homogenes E-Feld

```
\begin{tikzpicture}
  % Kondensatorplatten
  \draw[very thick] (0,0) -- (0,3) node[above] {$+$};
  \draw[very thick] (4,0) -- (4,3) node[above] {$-$};
  % Homogenes elektrisches Feld
  \foreach \y in {0.5,1,...,2.5} \draw[->] (0,\y) -- (4,
    \y);
  \node at (2,-0.4) {$\vec E$};
\end{tikzpicture}
```



Schwingung

```
\begin{tikzpicture}
  % Achsen
  \draw[->] (0,0) -- (7,0) node[right] {$t$};
  \draw[->] (0,-1.5) -- (0,1.5) node[above] {$y$};
  % Sinusschwingung
  \draw[thick, domain=0:6.5, samples=100] plot (\x, {sin
    (90*\x)});
  % Amplitude und Periodendauer
  \draw[<->] (1,0) -- node[right] {$\hat y$} (1,1);
  \draw[<->] (1,1.3) -- node[above] {$T$} (5,1.3);
\end{tikzpicture}
```



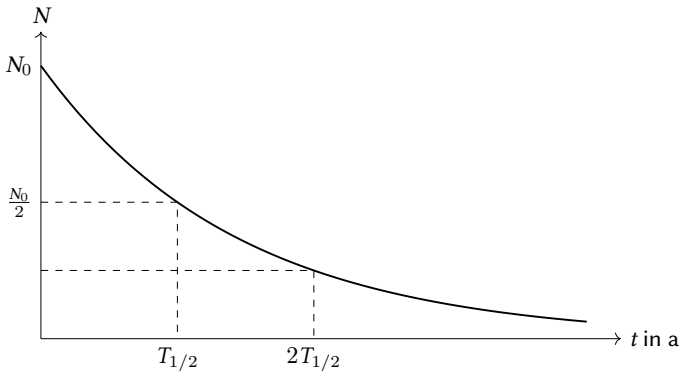
Zerfallsgleichung

```
\ce{^{238}_{92}U -> ^{234}_{90}Th + ^{4}_{2}He}
```

$$^{238}_{92}\mathrm{U} \longrightarrow {}^{234}_{90}\mathrm{Th} + {}^{4}_{2}\mathrm{He}$$

Zerfallskurve

```
\begin{tikzpicture}
  % Achsen
  \draw[->] (0,0) -- (8.5,0) node[right] {$t$ in a};
  \draw[->] (0,0) -- (0,4.5) node[above] {$N$};
  % Zerfallskurve
  \draw[thick, domain=0:8, samples=60] plot (\x, {4*exp
    (-0.3466*\x)});
  % Halbwertszeiten
  \draw[dashed] (0,2) -- (2,2) -- (2,0) node[below] {$T
    _{1/2}$};
  \draw[dashed] (0,1) -- (4,1) -- (4,0) node[below] {$2T
    _{1/2}$};
  \node[left] at (0,4) {$N_0$};
  \node[left] at (0,2) {$\frac{N_0}{2}$};
\end{tikzpicture}
```



Chemie

Summenformel

```
H$_2$O
```

H_2O

Reaktionsgleichung

```
\ce{2 H2 + O2 -> 2 H2O}
```

$2\,H_2 + O_2 \longrightarrow 2\,H_2O$

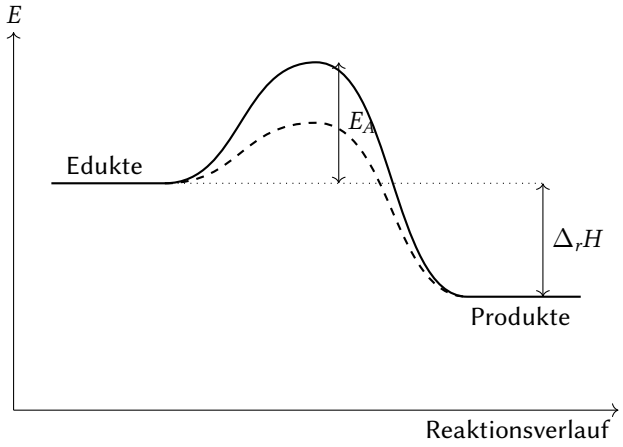
Lückenreaktion

```
\ce{Zn + 2 HCl ->} \LoesungLuecke{\ce{ZnCl2 + H2}}{4cm}
```

$Zn + 2\,HCl \longrightarrow \quad ZnCl_2 + H_2$

Energiediagramm

```
\begin{tikzpicture}
  % Achsen
  \draw[->] (0,0) -- (8,0) node[below left] {
    Reaktionsverlauf};
  \draw[->] (0,0) -- (0,5) node[above] {$E$};
  % Reaktionskurven (durchgezogen: unkatalysiert,
    gestrichelt: katalysiert)
  \draw[thick] (0.5,3) -- (2,3) .. controls (3,3) and
    (3,4.6) .. (4,4.6) .. controls (5,4.6) and (5,1.5) ..
     (6,1.5) -- (7.5,1.5);
  \draw[thick, dashed] (2,3) .. controls (3,3) and
    (3,3.8) .. (4,3.8) .. controls (5,3.8) and (5,1.5) ..
     (6,1.5);
  % Aktivierungsenergie und Reaktionsenthalpie
  \draw[dotted] (2,3) -- (7,3);
  \draw[<->] (4.3,3) -- (4.3,4.6) node[midway, right] {$E
    _A$};
  \draw[<->] (7,3) -- (7,1.5) node[midway, right] {$
    \Delta_r H$};
  % Edukte und Produkte
  \node[above] at (1.2,3) {Edukte};
  \node[below] at (6.7,1.5) {Produkte};
\end{tikzpicture}
```



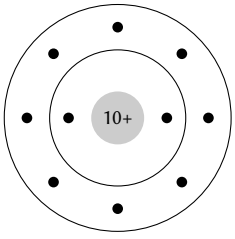
Schalenmodell

```
\begin{tikzpicture}
  % Atomkern
  \fill[gray!40] (0,0) circle (0.35);
  \node at (0,0) {\footnotesize 10+};

  % Schalengrenzen
  \draw (0,0) circle (0.9);
  \draw (0,0) circle (1.5);

  % Elektronen der K-Schale (im Raum zwischen Kern und 1.
    Grenze)
  \foreach \a in {0,180} \fill (\a:0.65) circle (2pt);

  % Elektronen der L-Schale (im Raum zwischen 1. und 2.
    Grenze)
  \foreach \a in {0,45,...,315} \fill (\a:1.2) circle (2
    pt);
\end{tikzpicture}
```



Elektronenkonfiguration

```
$1s^2\,2s^2\,2p^6$
```

$1s^2\,2s^2\,2p^6$

Periodensystem

```
\begin{tabular}{|c|c|c|c|c|c|c|c|}
  \hline
  I & II & III & IV & V & VI & VII & VIII \\ \hline
  H & & & & & & & He \\ \hline
  Li & Be & B & C & N & O & F & Ne \\ \hline
  Na & Mg & Al & Si & P & S & Cl & Ar \\ \hline
\end{tabular}
```

I	II	III	IV	V	VI	VII	VIII
H							He
Li	Be	B	C	N	O	F	Ne
Na	Mg	Al	Si	P	S	Cl	Ar

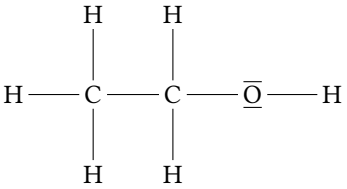
Ionengleichung

```
\ce{Ag+ + Cl- -> AgCl v}
```

$\ce{Ag+ + Cl- -> AgCl v}$

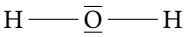
Strukturformel

```
\chemfig{H-C(-[2]H)(-[6]H)-C(-[2]H)(-[6]H)-\charge{90=
    \|,270=\|}{O}-H}
```



Lewis-Formel

```
\chemfig{H-\charge{90=\|,270=\|}{O}-H}
```



Redoxgleichungen

```
\begin{align*}
  \text{Oxidation:}\quad & \overset{0}{\ce{Cu}} \ce{->}
    \overset{+\mathrm{II}}{\ce{Cu^2+}} + 2\,\ce{e-} \\
  \text{Reduktion:}\quad & \overset{+\mathrm{I}}{\ce{Ag
    +}} + \ce{e-} \ce{->} \overset{0}{\ce{Ag}}
\end{align*}
```

$$\begin{aligned}\text{Oxidation:}\quad & \overset{0}{\ce{Cu}} \ce{->} \overset{+\mathrm{II}}{\ce{Cu^2+}} + 2\,\ce{e-} \\ \text{Reduktion:}\quad & \overset{+\mathrm{I}}{\ce{Ag+}} + \ce{e-} \ce{->} \overset{0}{\ce{Ag}}\end{aligned}$$

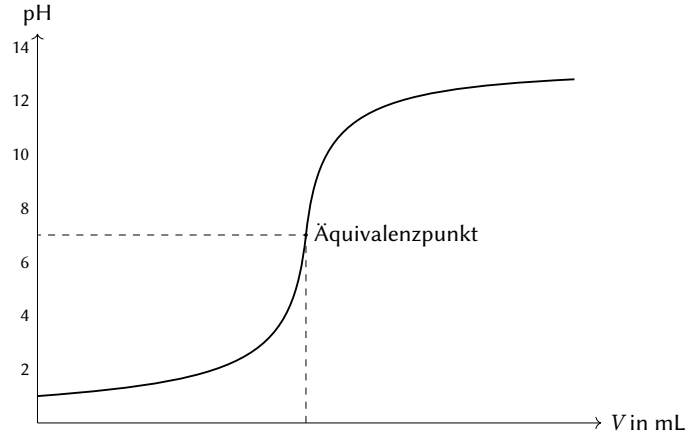
Gleichgewicht

```
\ce{N2 + 3 H2 <=> 2 NH3}
```

$\ce{N2 + 3 H2 <=> 2 NH3}$

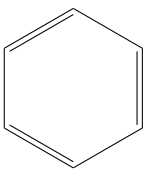
Titrationskurve

```
\begin{tikzpicture}[scale=0.4]
  % Achsen
  \draw[->] (0,0) -- (21,0) node[right] {$V$ in mL};
  \draw[->] (0,0) -- (0,14.5) node[above] {pH};
  \foreach \y in {2,4,...,14} \node[left] at (0,\y) {
    \footnotesize \y};
  % Titrationskurve
  \draw[thick] (0,1) .. controls (8,1.6) and (9.5,2.5) ..
     (10,7) .. controls (10.5,11.5) and (12,12.4) ..
    (20,12.8);
  % Aequivalenzpunkt
  \draw[dashed] (10,0) -- (10,7) -- (0,7);
  \fill (10,7) circle (2pt) node[right] {Äquivalenzpunkt
    };
\end{tikzpicture}
```



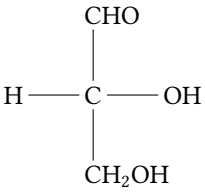
Benzolring

```
\chemfig{*6(=-=-=-)}
```



Fischer-Projektion

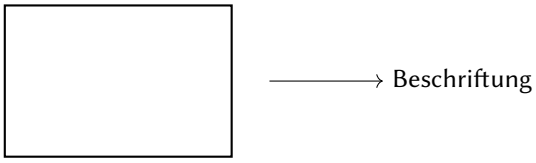
```
\chemfig{CHO-[6]C(-[4]H)(-[0]OH)-[6]CH_2OH}
```



Biologie

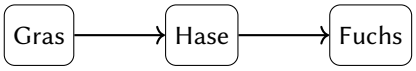
Beschriftete Skizze

```
\begin{tikzpicture}
  % Zeichenrahmen
  \draw[thick] (0,0) rectangle (3,2);
  % Beschriftungslinie
  \draw[->] (3.5,1) -- (5,1) node[right] {Beschriftung};
\end{tikzpicture}
```



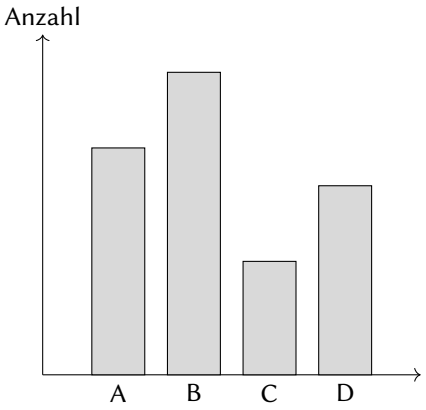
Nahrungskette

```
\begin{tikzpicture}[node distance=1.2cm, box/.style={draw
    , rounded corners, minimum height=0.8cm}]
  % Nahrungskette: Produzent und Konsumenten
  \node[box] (a) {Gras};
  \node[box, right=of a] (b) {Hase};
  \node[box, right=of b] (c) {Fuchs};
  % Nahrungsfluss
  \draw[->, thick] (a) -- (b);
  \draw[->, thick] (b) -- (c);
\end{tikzpicture}
```



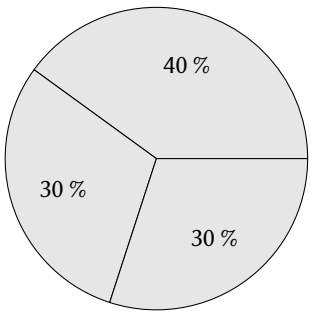
Säulendiagramm

```
\begin{tikzpicture}
  % Achsen
  \draw[->] (0,0) -- (5,0);
  \draw[->] (0,0) -- (0,4.5) node[above] {Anzahl};
  % Datenbalken
  \foreach \x/\h/\l in {1/3/A, 2/4/B, 3/1.5/C, 4/2.5/D} {
    \draw[fill=gray!30] (\x-0.35,0) rectangle (\x+0.35,\h
    );
    \node[below] at (\x,0) {\l};
  }
\end{tikzpicture}
```



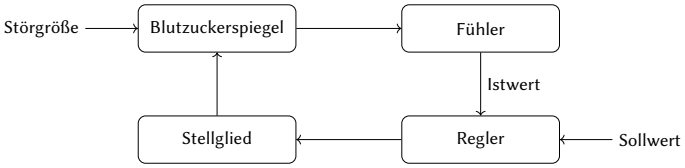
Kreisdiagramm

```
\begin{tikzpicture}
  % Kreissektoren: Startwinkel / Endwinkel / Prozentwert
  \foreach \a/\b/\l in {0/144/40\,\%, 144/252/30\,\%,
     252/360/30\,\%} {
    \draw[fill=gray!20] (0,0) -- (\a:2) arc (\a:\b:2) --
    cycle;
    \node at ({(\a+\b)/2}:1.3) {\l};
  }
\end{tikzpicture}
```



Regelkreis

```
\begin{tikzpicture}[node distance=1.2cm, box/.style={draw
    , rounded corners, minimum width=3cm, minimum height
    =0.9cm, align=center}]
  % Regelkreis-Glieder
  \node[box] (ist) {Blutzuckerspiegel};
  \node[box, right=2cm of ist] (f) {Fühler};
  \node[box, below=of f] (r) {Regler};
  \node[box, below=of ist] (s) {Stellglied};
  % Signalfluss im Regelkreis
  \draw[->] (ist) -- (f);
  \draw[->] (f) -- node[right] {Istwert} (r);
  \draw[->] (r) -- (s);
  \draw[->] (s) -- (ist);
  % Externe Einfluesse (Sollwert und Stoergroesse)
  \draw[<-] (r.east) -- ++(1,0) node[right] {Sollwert};
  \draw[<-] (ist.west) -- ++(-1,0) node[left] {Störgröße
    };
\end{tikzpicture}
```



Basensequenz

```
\begin{tabular}{l l}
  DNA (codogener Strang): & \texttt{TAC GGA TTC} \\
  mRNA: & \LoesungLuecke{\texttt{AUG CCU AAG}}{4cm} \\
  Aminosäuren: & \LoesungLuecke{Met -- Pro -- Lys}{4cm}
\end{tabular}
```

DNA (codogener Strang):	TAC GGA TTC
mRNA:	AUG CCU AAG
Aminosäuren:	Met – Pro – Lys

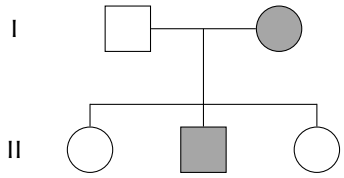
Kreuzungsschema

```
{\renewcommand{\arraystretch}{1.6}\begin{tabular}{c|>{
    \centering\arraybackslash}p{1.5cm}|>{\centering
    \arraybackslash}p{1.5cm}}
   & $A$ & $a$ \\ \hline
  $A$ & \LoesungLeer{$AA$}{0pt} & \LoesungLeer{$Aa$}{0pt}
    \\ \hline
  $a$ & \LoesungLeer{$Aa$}{0pt} & \LoesungLeer{$aa$}{0pt}
\end{tabular}}
```

	A	a
A	AA	Aa
a	Aa	aa

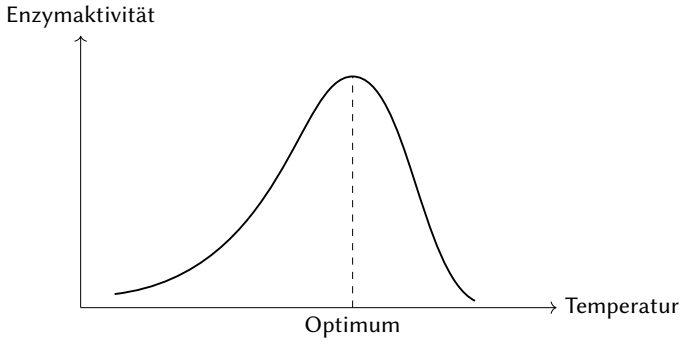
Stammbaum

```
\begin{tikzpicture}[male/.style={draw, minimum size=6mm},
    female/.style={draw, circle, minimum size=6mm},
   affected/.style={fill=gray!70}]
  % Generationen
  \node at (-1.5,0) {I};
  \node at (-1.5,-1.6) {II};
  % Eltern
  \node[male] (v) at (0,0) {};
  \node[female, affected] (m) at (2,0) {};
  \draw (v) -- (m);
  \draw (1,0) -- (1,-1);
  \draw (-0.5,-1) -- (2.5,-1);
  % Kinder
  \node[female] (k1) at (-0.5,-1.6) {};
  \node[male, affected] (k2) at (1,-1.6) {};
  \node[female] (k3) at (2.5,-1.6) {};
  \foreach \k in {k1,k2,k3} \draw (\k.north) -- ++(0,0.3)
    ;
\end{tikzpicture}
```



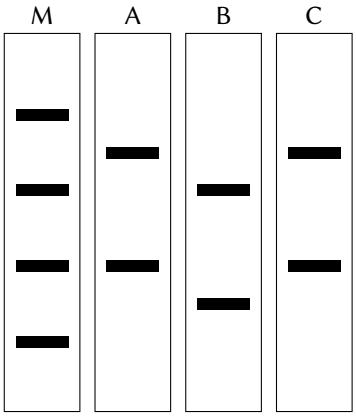
Enzymaktivität

```
\begin{tikzpicture}
  % Achsen
  \draw[->] (0,0) -- (7,0) node[right] {Temperatur};
  \draw[->] (0,0) -- (0,4) node[above] {Enzymaktivität};
  % Aktivitaetskurve
  \draw[thick] (0.5,0.2) .. controls (3,0.5) and
    (3.2,3.4) .. (4,3.4) .. controls (4.8,3.4) and
    (5,0.5) .. (5.8,0.1);
  % Temperaturoptimum
  \draw[dashed] (4,0) node[below] {Optimum} -- (4,3.4);
\end{tikzpicture}
```



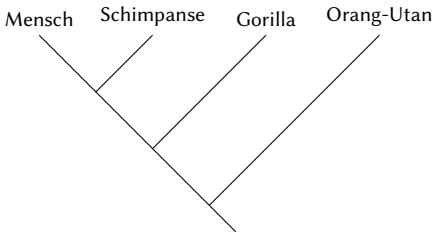
Gelelektrophorese

```
\begin{tikzpicture}
  % Geltaschen und Bahnenbeschriftung
  \foreach \i/\l in {0/M, 1/A, 2/B, 3/C} {
    \draw (\i*1.2,0) rectangle ++(1,-5);
    \node[above] at (\i*1.2+0.5,0) {\l};
  }
  % Bandenmuster
  \foreach \i/\y in {0/-1, 0/-2, 0/-3, 0/-4, 1/-1.5,
     1/-3, 2/-2, 2/-3.5, 3/-1.5, 3/-3}
    \fill (\i*1.2+0.15,\y) rectangle ++(0.7,-0.15);
\end{tikzpicture}
```



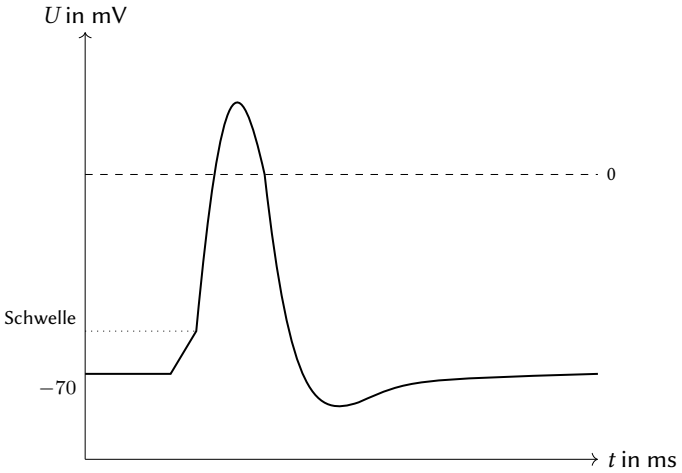
Kladogramm

```
\begin{tikzpicture}
  % Kladogramm-Verzweigungen
  \draw (0,3) -- (0.75,2.25) -- (1.5,3);
  \draw (0.75,2.25) -- (1.5,1.5) -- (3,3);
  \draw (1.5,1.5) -- (2.25,0.75) -- (4.5,3);
  \draw (2.25,0.75) -- (2.6,0.4);
  % Taxa
  \foreach \x/\n in {0/Mensch, 1.5/Schimpanse, 3/Gorilla,
     4.5/Orang-Utan}
    \node[above, font=\footnotesize] at (\x,3) {\n};
\end{tikzpicture}
```



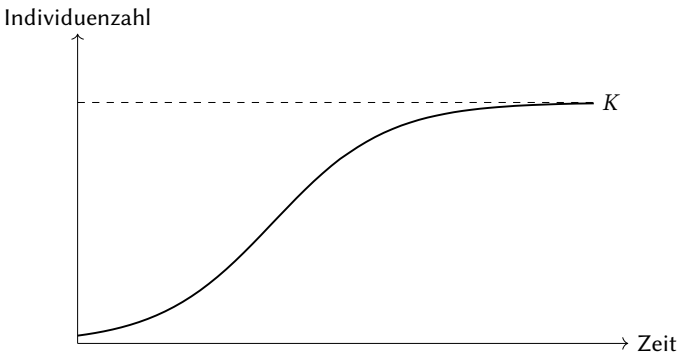
Aktionspotenzial

```
\begin{tikzpicture}[xscale=1.2, yscale=0.04]
  % Achsen
  \draw[->] (0,-100) -- (0,50) node[above] {$U$ in mV};
  \draw[->] (0,-100) -- (6,-100) node[right] {$t$ in ms};
  % Nulllinie und Schwellenwert
  \draw[dashed] (0,0) -- (6,0) node[right] {\footnotesize
     0};
  \draw[dotted] (0,-55) -- (1.3,-55);
  % Aktionspotenzial-Verlauf
  \draw[thick] (0,-70) -- (1,-70) -- (1.3,-55) ..
    controls (1.6,40) and (1.8,40) .. (2.1,0)
    .. controls (2.4,-80) and (2.7,-85) .. (3.2,-80) ..
    controls (3.8,-72) .. (6,-70);
  \node[left] at (0,-75) {\footnotesize $-70$};
  \node[left] at (0,-50) {\footnotesize Schwelle};
\end{tikzpicture}
```



Populationswachstum

```
\begin{tikzpicture}
  % Achsen
  \draw[->] (0,0) -- (8,0) node[right] {Zeit};
  \draw[->] (0,0) -- (0,4.5) node[above] {Individuenzahl
    };
  % Umweltkapazitaet K
  \draw[dashed] (0,3.5) -- (7.5,3.5) node[right] {$K$};
  % Logistisches Wachstum
  \draw[thick, domain=0:7.5, samples=60] plot (\x,
    {3.5/(1+30*exp(-1.2*\x))});
\end{tikzpicture}
```



Sprachen & Gesellschaft

Quellentext

```
\begin{lstlisting}[language={}, numbers=left, stepnumber
    =5, firstnumber=1, basicstyle=\normalfont, columns=
    fullflexible, keepspaces]
Text der Quelle
\end{lstlisting}
\begin{flushright}\footnotesize Quelle: Autor, Titel,
    Jahr\end{flushright}
```

Text der Quelle

Quelle: Autor, Titel, Jahr

Lückentext

```
Der Hund \LoesungLuecke{bellt}{2cm} laut, weil er
    \LoesungLuecke{Hunger}{2cm} hat.
```

Der Hund bellt laut, weil er Hunger hat.

Zuordnung

```
\begin{tabular}{ll@{\hspace{2cm}}ll}
  1 & Begriff & A & Erklärung \\
  2 & Begriff & B & Erklärung \\
  3 & Begriff & C & Erklärung
\end{tabular}

{\renewcommand{\arraystretch}{1.6}\begin{tabular
    }{|*{3}{>{\centering\arraybackslash}p{1cm}|}}
  \hline
  1 & 2 & 3 \\ \hline
  \LoesungLeer{B}{0pt} & \LoesungLeer{C}{0pt} &
    \LoesungLeer{A}{0pt} \\ \hline
\end{tabular}}
```

1	Begriff		A	Erklärung
2	Begriff		B	Erklärung
3	Begriff		C	Erklärung

1	2	3
B	C	A

Vokabeltabelle

```
{\renewcommand{\arraystretch}{1.6}\begin{tabular}{|l|p{5
    cm}|}
  \hline
  \textbf{Deutsch} & \textbf{Englisch} \\ \hline
  der Hund & \LoesungLeer{the dog}{0pt} \\ \hline
  die Katze & \LoesungLeer{the cat}{0pt} \\ \hline
\end{tabular}}
```

Deutsch	**Englisch**
der Hund	the dog
die Katze	the cat

Lückentabelle

```
{\renewcommand{\arraystretch}{1.6}\begin{tabular}{|l|p{6
    cm}|}
  \hline
  Begriff & \LoesungLeer{Antwort}{0pt} \\ \hline
  Begriff & \LoesungLeer{Antwort}{0pt} \\ \hline
\end{tabular}}
```

Begriff	Antwort
Begriff	Antwort

Zeitleiste

```
\begin{tikzpicture}
  % Zeitstrahl
  \draw[->, thick] (0,0) -- (12,0);
  % Markierungen und Ereignisse
  \foreach \x/\j/\e in {1/1789/Französische Revolution,
    5/1871/Reichsgründung, 9/1918/Ende des 1.\ Weltkriegs
    } {
    \draw (\x,0.15) -- (\x,-0.15) node[below] {\j};
    \node[above, align=center, text width=2.5cm] at (\x
    ,0.2) {\footnotesize \e};
  }
\end{tikzpicture}
```

